

WORKED SOLUTION · ONE OF MANY VALID ANSWERS

Insurance Policy Management System

Applying the seven product-engineering modules, step by step

- 1 Being a product engineer
- 2 Quality-first thinking
- 3 Business outcomes
- 4 Value stream focus
- 5 User experience
- 6 Speed and agility
- 7 DevOps practices to scale

Exercise brief: 4 epics · 45 minutes · teams of 3

STEP 0 · READ THE BRIEF

The system at a glance

Before applying any module, agree on the actors, the four epics and how the 45 minutes will be spent



Epic 1 · Policy discovery & application

Compare products · Apply online · Instant quote



Epic 2 · Risk assessment

Integrated data · AI/ML risk model · IoT monitoring



Epic 3 · Claims processing & payment

File with documents · Review & approve · Settlement



Epic 4 · Insurance admin dashboard

Policy performance · Claims analytics · Regulatory reporting

WHO IS IN THE SYSTEM

Policyholder (Meera, 34) — wants cover fast, hates paperwork, fears claim rejection

Underwriter (Rajan) — wants clean risk data and fewer manual referrals

Claims adjuster (Priya) — wants complete documents and a clear approval queue

Ops / compliance head (Vikram) — wants loss-ratio visibility and audit-ready IRDAI reports

45-MINUTE PLAN

5 min read brief, name actors, pick a North Star metric

25 min modules 1–7, ~3–4 min each, one epic in depth per module

10 min fill the 7 × 4 matrix (last slide) and pick your top 3 bets

5 min present: one insight per module, 30 seconds each



Team checkpoint Which ONE number would tell you in 90 days whether this system is working? Write it down now — every module will refer back to it.

What it means to be a product engineer

Own the outcome, not the ticket. Reframe every epic from “build the feature” to “solve the customer’s problem and prove it”

Epic	Feature-factory framing	Product-engineer framing	What you would do differently
1 · Discovery & application	“Build a comparison page and an application form with instant quote API.”	“Meera abandons at step 4 of 9 because we ask for RC-book details she doesn’t have. Get her to a bindable quote in under 3 minutes.”	Instrument the funnel first; pre-fill from vehicle registry / Aadhaar-verified data; quote before full KYC
2 · Risk assessment	“Train a risk model with 0.85 AUC and expose /score.”	“Underwriters override 40% of scores. A model they don’t trust is a model that isn’t used.”	Ship explanations with every score; track override rate as a product metric; pair with Rajan weekly
3 · Claims & payment	“Upload documents, approval workflow, payout integration.”	“Claim rejection is the #1 reason customers leave. Make status transparent and settle clean claims in 48 hours.”	Define ‘clean claim’ and auto-approve it; show a live status timeline; measure time-to-settle, not tickets closed
4 · Admin dashboard	“Dashboards for policies, claims, and a regulatory export.”	“Vikram needs to spot a deteriorating loss ratio by segment before quarter-end, and file IRDAI returns without a spreadsheet marathon.”	Start from the 3 decisions the dashboard must enable; alerts over charts; reports generated, not assembled

 **Team checkpoint** Can each of you say, for one epic, what problem it solves, for whom, and how you will know? If the answer is a feature name, you are still in the factory.

Quality-first thinking

Quality is defined per epic, designed in from the first commit, and verified continuously — not inspected at the end



Epic 1 • Policy discovery & application

Quality means: Correctness of money & terms

A quote must equal the bound premium to the paisa; product rules (exclusions, add-ons) encoded once, tested as a rule table.

Build it in: Contract tests on the quote engine · golden-file tests for 200 product/rider combos · property-based tests on premium maths



Epic 2 • Risk assessment

Quality means: Trustworthy, fair, explainable

Score drift, biased features (pin-code as proxy), and stale IoT feeds silently corrupt underwriting.

Build it in: Model validation suite + fairness checks in CI · data-quality gates on every feed · shadow-mode before any model goes live



Epic 3 • Claims processing & payment

Quality means: Nothing lost, nothing double-paid

Documents, approvals and payouts are a chain of custody; a lost file or a duplicate settlement is a regulatory incident.

Build it in: Idempotent payout API · event-sourced claim history · reconciliation job vs bank statements · fraud rules as tested code



Epic 4 • Insurance admin dashboard

Quality means: Numbers the CFO can sign

Every figure feeds board packs and IRDAI filings; a wrong loss ratio is worse than a late one.

Build it in: Single metrics layer (one definition of 'earned premium') · reconciled nightly to the ledger · report snapshots immutable



Team checkpoint What is the one defect that would get this system on the front page? Design the test for that first, then the happy path.

Business outcomes

Connect every epic to money the insurer makes or loses — then choose leading indicators the team can move this sprint

 **North Star: profitable growth** — Gross Written Premium $\uparrow$ while Combined Ratio (claims + expenses $\div$ premium) stays < 100%. Every epic must move one side of that fraction or shorten the time to see it.

Epic	Business outcome it drives	Leading metric (weekly)	Lagging metric (quarterly)	Guardrail (must not break)
1 · Discovery & application	More premium per marketing rupee	Quote-to-bind conversion · median time to quote · form abandonment step	Cost of acquisition · GWP from digital channel	Mis-selling complaints · KYC rejections after bind
2 · Risk assessment	Price risk correctly $\rightarrow$ lower loss ratio without losing good customers	Straight-through-processing rate · underwriter override rate · IoT-discount uptake	Loss ratio by segment vs priced expectation	Fairness disparity across cohorts · model drift alerts
3 · Claims & payment	Retention and lower leakage & expense per claim	Time to first response · % clean claims auto-approved · docs requested per claim	Renewal rate after a claim · claims expense ratio · fraud leakage	Wrongful-rejection appeals · duplicate payouts
4 · Admin dashboard	Faster, better portfolio decisions; zero regulatory penalties	Decisions logged from dashboard alerts · report prep hours	Filing timeliness · reserve accuracy · underwriting actions taken	Metric definitions diverging from finance ledger

 **Team checkpoint** For your North Star from Step 0: which epic moves it fastest, and which metric would you show the CEO in week 2 to prove it is moving?

Value stream focus

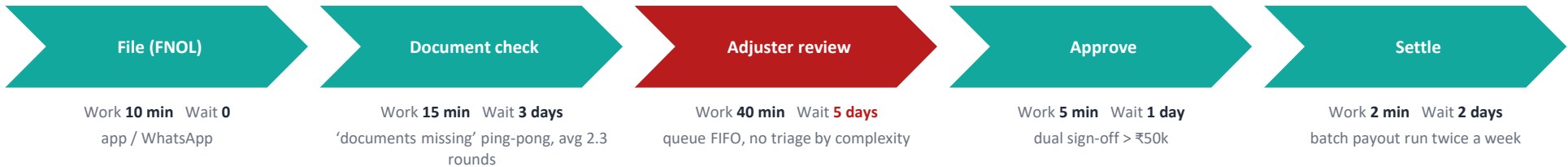
Map the two flows customers actually feel — quote-to-bind and claim-to-cash — and attack the biggest wait, not the busiest step

VALUE STREAM A · Quote-to-bind (Epics 1 + 2)



Flow efficiency ≈ 20 min work in ~ 4 days = $< 1\%$. Biggest lever: cut manual referrals (explainable score + auto-approve low-risk bands) and pre-fill from registries. Target: 80% of quotes bound same session.

VALUE STREAM B · Claim-to-cash (Epic 3, fed by Epic 2 data)



≈ 72 min of work in ~ 11 days. Biggest lever: triage by complexity (clean claims $\rightarrow$ auto-approve $\rightarrow$ same-day payout) and a document checklist at FNOL. Target: clean claims settled in 48 h; this is also the retention driver from Module 3.

Team checkpoint Which single wait, if removed, would change a customer's story about you? That is your first slice in Module 6 — not the step where engineers spend the most time.

User experience

Four users, four very different moments of truth — design for the anxious moment, not the average screen

Epic · primary user	Moment that matters	What they feel	UX principle	Concrete design decision
1 · Meera, buying cover	Comparing 6 plans and being asked to fill a 9-step form	Overwhelmed; suspicious of fine print	Progressive disclosure · plain-language exclusions · quote first, KYC later	3 plans max, 'what's NOT covered' shown up front; save-and-resume via OTP; instant indicative quote after 4 fields
2 · Rajan, underwriting	A risk score lands in his queue with no reason	Distrust; reverts to gut feel	Explainability · human-in-the-loop by design · consistency	Top-3 factors and comparable cases beside every score; one-click 'accept / override with reason' that feeds the model back
3 · Meera, filing a claim	Day 4 after the accident, no update, hospital asking for money	Anxious, powerless	Transparency · proactive communication · reduce effort at the worst time	Live status timeline with named next step & ETA; photo-guided document capture; WhatsApp updates; 'what happens next' on every screen
4 · Vikram, running the book	Board asks why loss ratio jumped in two-wheeler, Tier-2 cities	Blind; drowning in exports	Decision-first dashboards · alerts over charts · trust through lineage	Exception-driven home page (what changed, why it matters); drill-down to policy level; every number shows its source & refresh time

 **Team checkpoint** Pick one user and one moment. Sketch the single screen they see at that moment. Does it answer “what is happening, what happens next, what do I need to do?”

Speed and agility

Slice each epic into thin, end-to-end, releasable increments — learn from real users every two weeks instead of demoing at month nine

Epic	Slice 1 · walking skeleton (weeks 1–2)	Slice 2 · learn & extend (weeks 3–6)	Slice 3 · scale (weeks 7+)	Deliberately NOT yet
1 · Discovery & application	One product (two-wheeler), 4-field indicative quote, apply via existing form, manual bind	Add comparison of 3 plans; save & resume; pre-fill from vehicle registry	All lines of business; partner/agent channel; A/B pricing presentation	Full product catalogue, chatbot, multi-language
2 · Risk assessment	Rules-based score with 5 factors, shown to underwriters in shadow mode	ML model in shadow; explanations; auto-approve lowest-risk band only	IoT telematics discount pilot with one device partner; retraining pipeline	Real-time IoT pricing, external data marketplace
3 · Claims & payment	FNOL via app with photo upload; status timeline; manual review as today	Document checklist & auto-validation; triage clean vs complex; daily payout run	Auto-approval of clean claims < ₹25k; fraud scoring; instant payout	Cashless garage network integration, subrogation
4 · Admin dashboard	One page: loss ratio & claims aging by product, refreshed nightly from the ledger	Alerts on threshold breaches; drill-down; one IRDAI return generated end-to-end	Self-serve segments, reserve modelling, all statutory returns	Predictive portfolio simulation, BI for every team

Feature flags

every slice dark-launched, on for 5% first

Two-week release train

no slice bigger than a train

Hypothesis per slice

“we believe... we’ll know when...”

Kill criteria

decide up front what result stops a slice



Team checkpoint Is your Slice 1 something a real policyholder or underwriter could use in two weeks? If it is a ‘backend first’ plan, it is a phase, not a slice.

DevOps practices to scale

The platform that lets four epics ship independently, safely, and under regulatory scrutiny



CI/CD per service

Trunk-based; every commit runs unit + contract + rule-table tests; deploy to prod behind flags in < 1 h. Quote engine and claims service release independently.



Infrastructure & environments as code

Ephemeral preview environments per PR; prod-like data via masked synthetic policies; DR region rehearsed quarterly (regulatory expectation).



Observability tied to outcomes

Traces across quote → score → bind; business dashboards (conversion, STP rate, time-to-settle) sit next to latency & errors; SLOs per value stream.



MLOps for the risk model

Versioned features & models; shadow → canary → full; drift and fairness monitors; one-click rollback; every score logged with model version for audit.



Streaming & IoT data platform

Event backbone (policy, claim, telemetry events); schema registry; replayable streams so analytics and the dashboard read the same truth as the transactions.



Security & compliance as code

Policy-as-code (data residency, PII masking, retention); audit trail generated from events; access reviews automated; IRDAI report reproducible from immutable snapshots.

daily per service

Deploy frequency

< 1 day

Lead time for change

< 10%

Change failure rate

< 1 h

Time to restore

< 15 min

+ Model rollback

? Team checkpoint Which practice, if missing, would make Module 6's two-week slices impossible or unsafe here? Fund that one before any epic.

PUTTING IT TOGETHER

The 7 × 4 matrix — one line per cell

Fill this in for your own answer; then circle the three cells you would fund first

Module ↓ · Epic →	1 · Policy discovery & application	2 · Risk assessment	3 · Claims processing & payment	4 · Insurance admin dashboard
1 · Product engineer	Time-to-bindable-quote, not a form	Explainable, trusted scores	48-h clean-claim promise	Decisions, not dashboards
2 · Quality-first	Premium maths golden tests	Fairness & drift gates in CI	Idempotent, reconciled payouts	One metrics layer, ledger-reconciled
3 · Business outcomes	Quote-to-bind conversion	STP rate, loss ratio by segment	Time-to-settle, post-claim renewal	Filing timeliness, decisions logged
4 · Value stream	Kill the manual-referral wait	Feeds both streams with data	Triage clean vs complex at FNOL	Surfaces flow metrics
5 · User experience	Quote after 4 fields, exclusions up front	Top-3 reasons beside every score	Live timeline + next step	Exception-driven home page
6 · Speed & agility	Two-wheeler, 4-field quote first	Rules in shadow → ML → IoT	FNOL + timeline first; auto-approve later	One page, one return, nightly
7 · DevOps to scale	Independent quote-engine pipeline	MLOps: shadow/canary/rollback	Event-sourced claims + payout SLOs	Immutable snapshots, policy-as-code



What a strong team answer looks like

Names a North Star and traces every module back to it · treats underwriters and ops as users, not stakeholders · finds the wait, not the work, in the value stream · Slice 1 is usable by a real person in two weeks · quality and compliance are designed in, not bolted on · disagrees with this model answer somewhere, with a reason.